



Turing Machines

Computation and Discrete Structures III – Unit 3: Turing Machines and Applications

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Material adapted from Professor Oscar Bedoya

2026

1. Turing Machines

2. Designing Turing Machines

1. Turing Machines

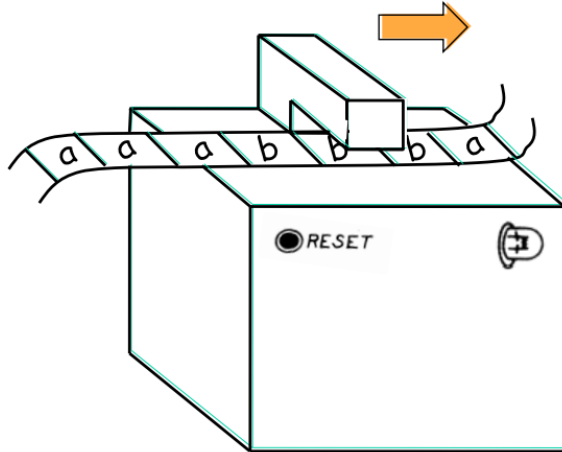
2. Designing Turing Machines

Alan Turing

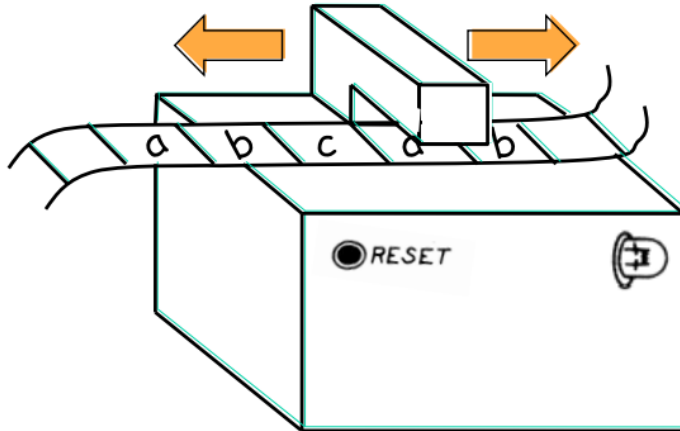
- English mathematician, computer scientist, and cryptographer
- Proposed the Turing Test
- Creator of the Turing machine, inspired by the operation of a typewriter



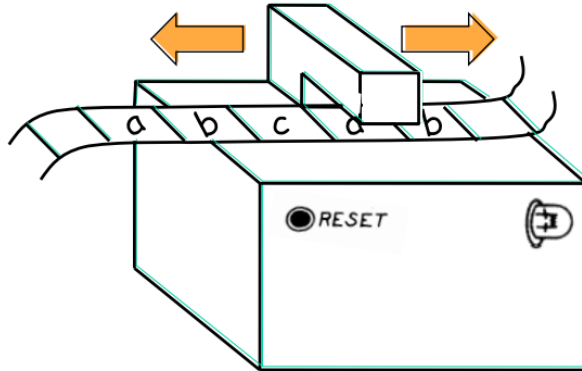
Finite Automaton



Turing Machine

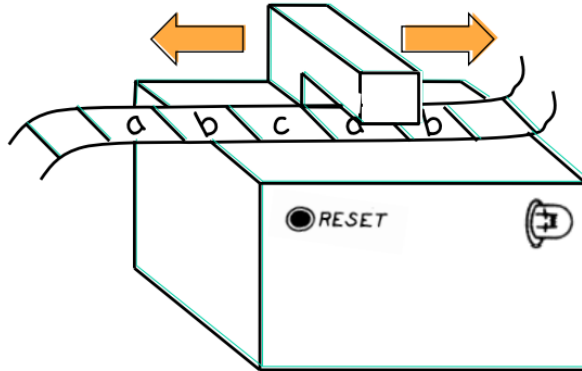


Turing Machine



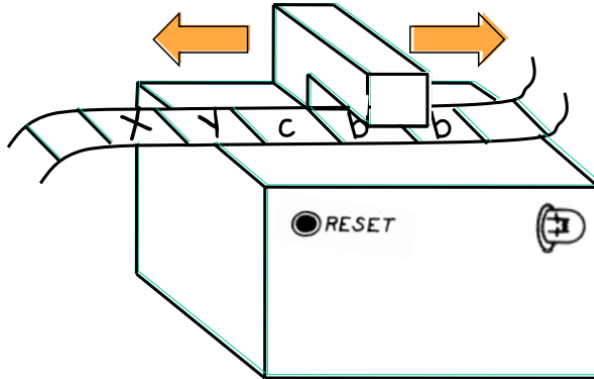
- The head above the tape is a read/write head and can move in both directions

Turing Machine

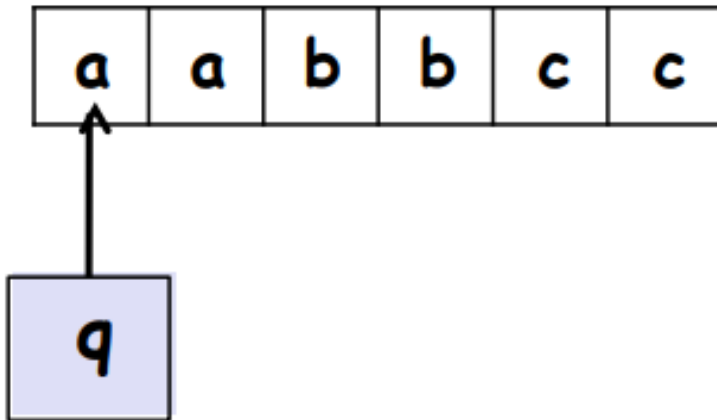


- It is written to the same tape using the input string

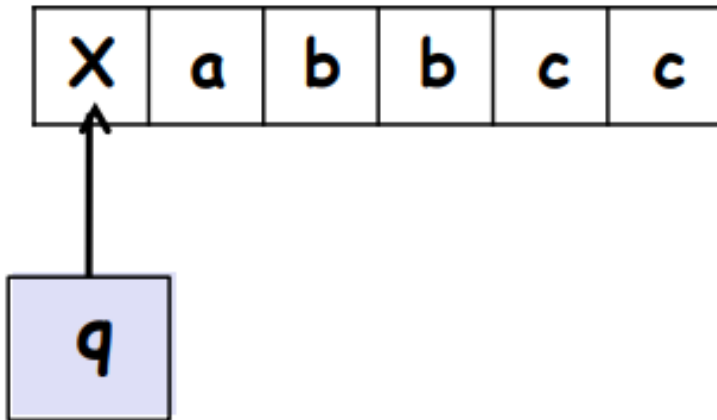
Turing Machine



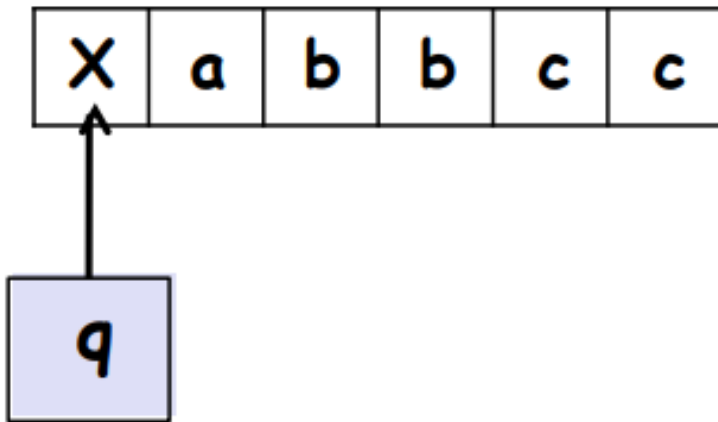
- It is written to the same tape using the input string



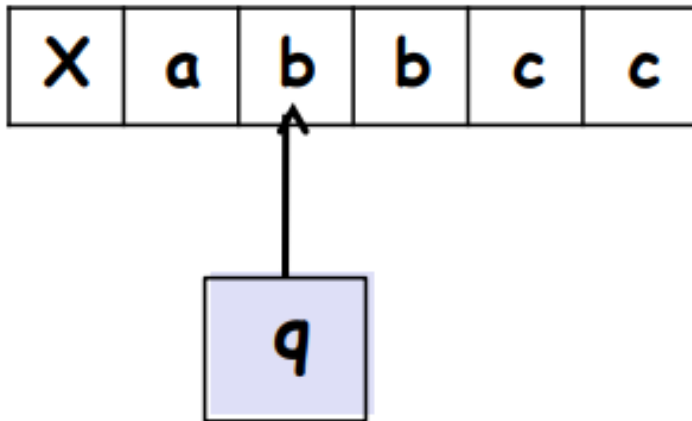
Writing on the tape



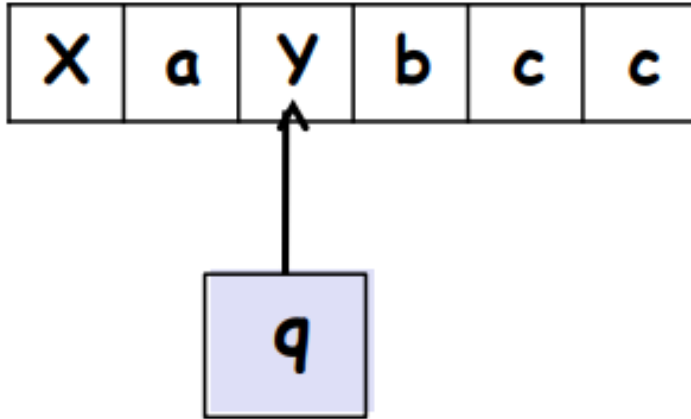
Writing on the tape



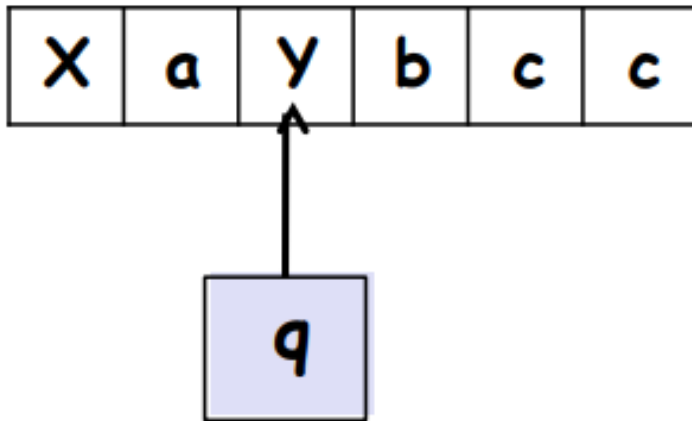
*Moving along the tape
searching for a b*



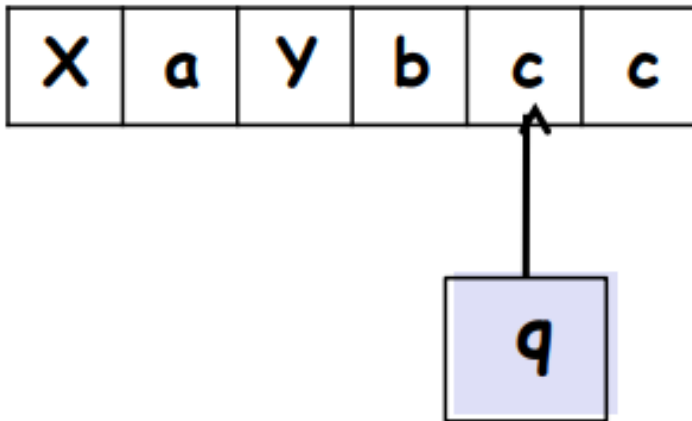
*Moving along the tape
searching for a b*



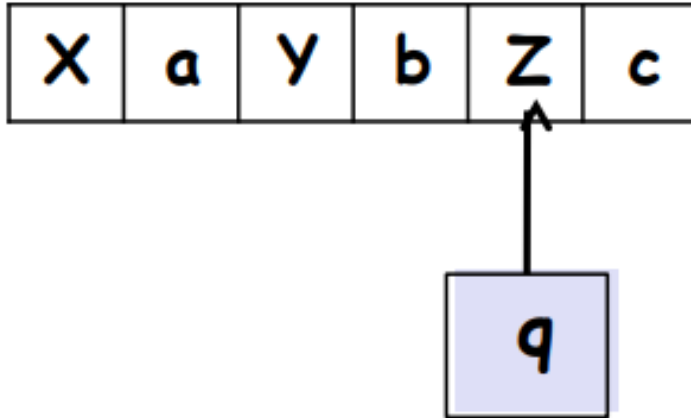
Replacing it with Y



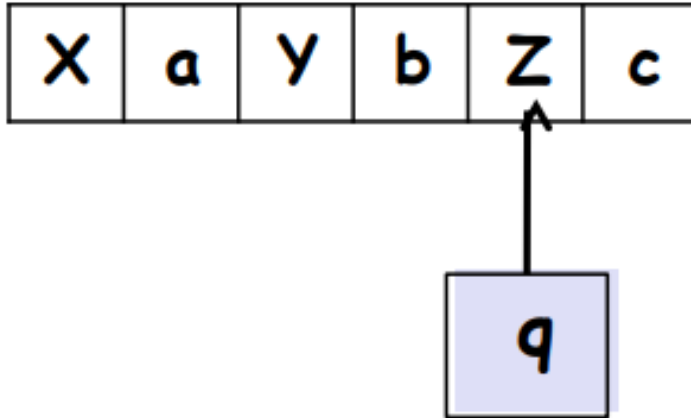
*Moving along the tape
searching for a c*



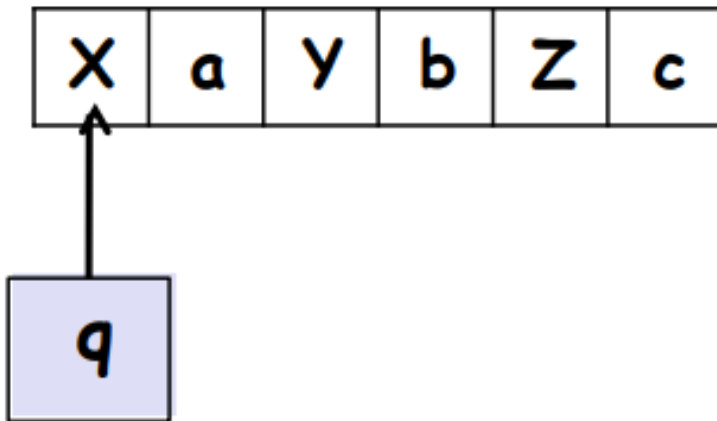
*Moving along the tape
searching for a c*



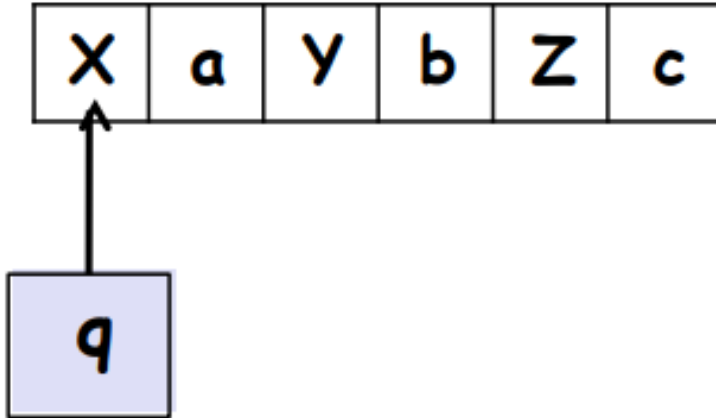
Replacing it with Z



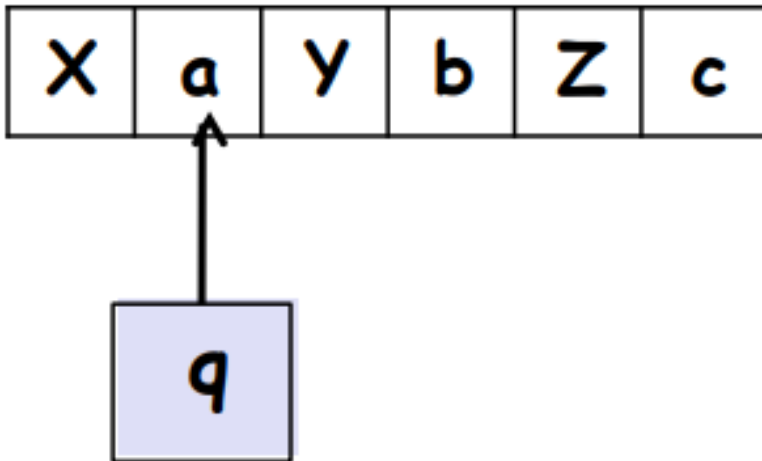
Moving left until finding an X

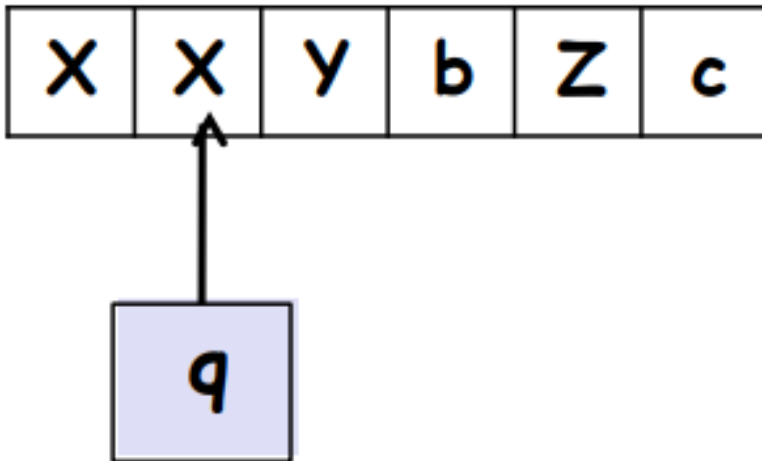


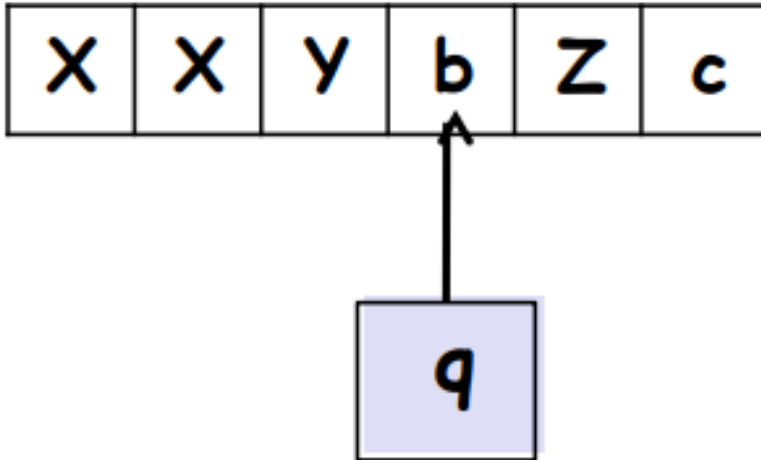
Moving left until finding an X

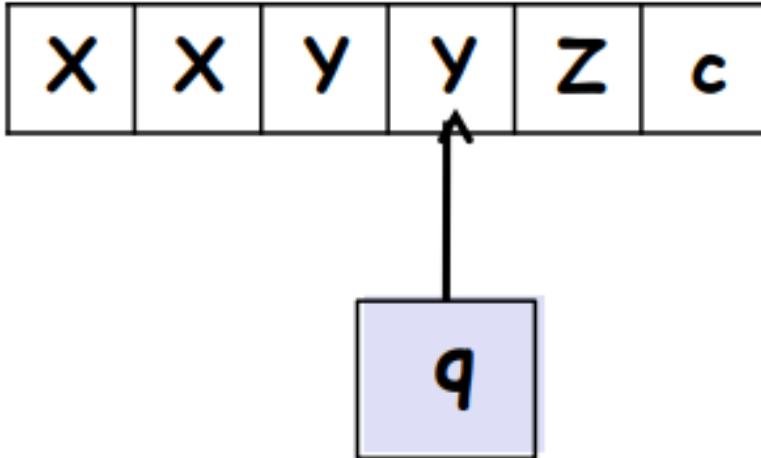


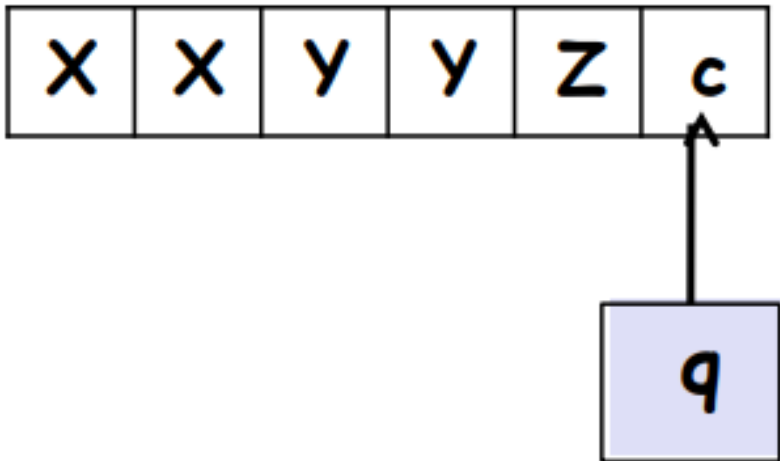
Repeating the procedure

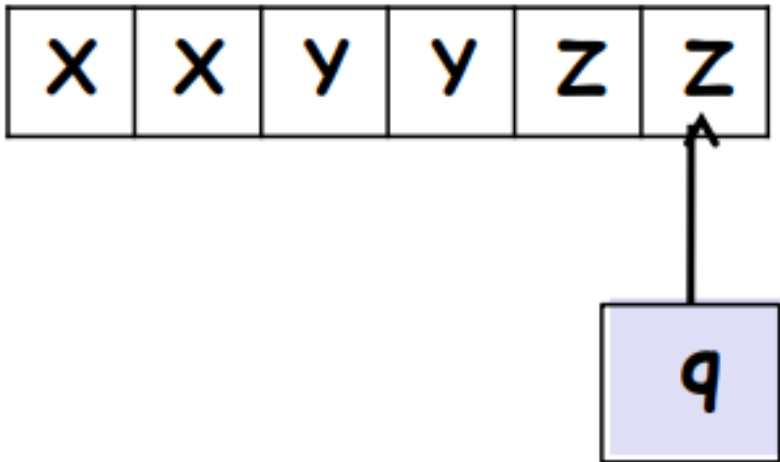


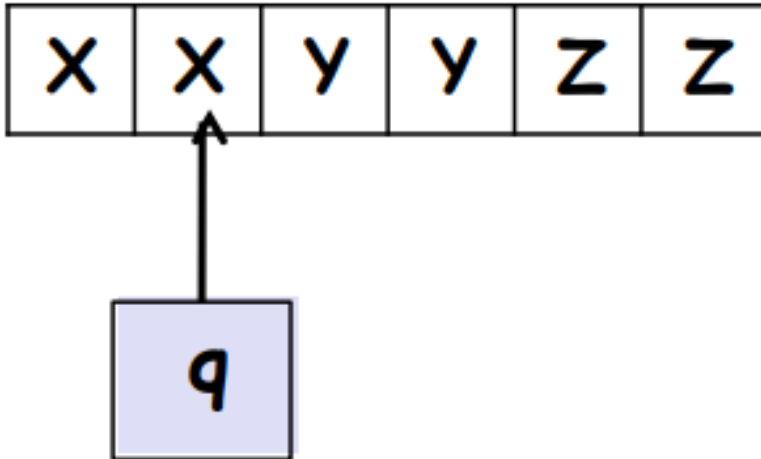


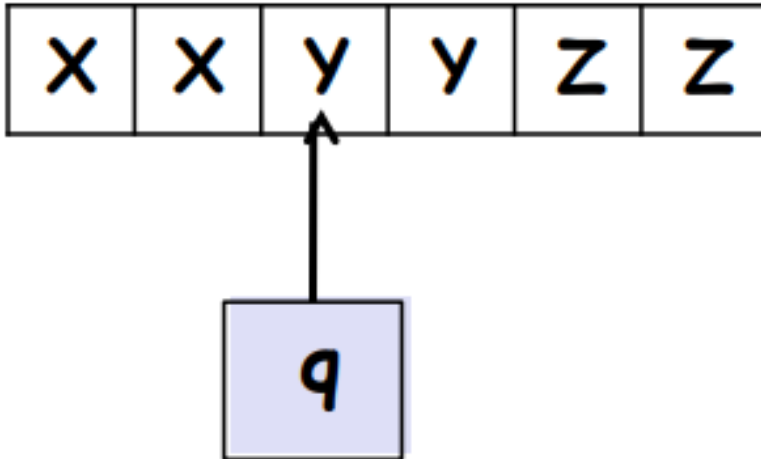


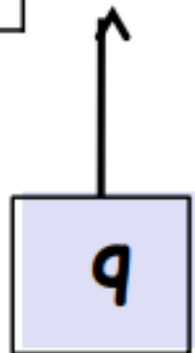
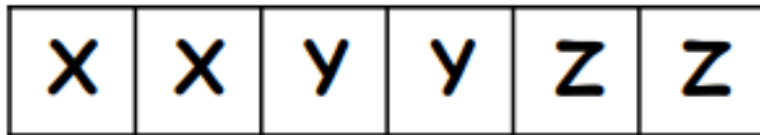


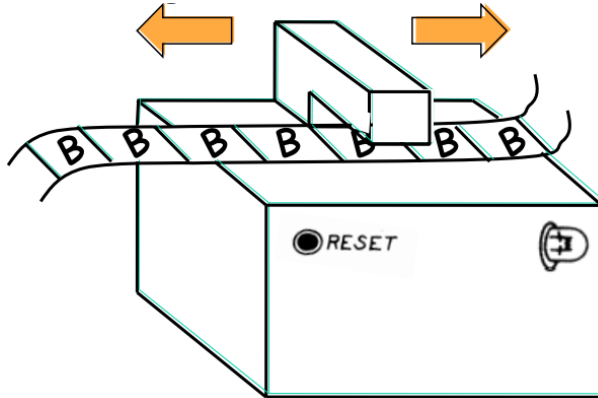




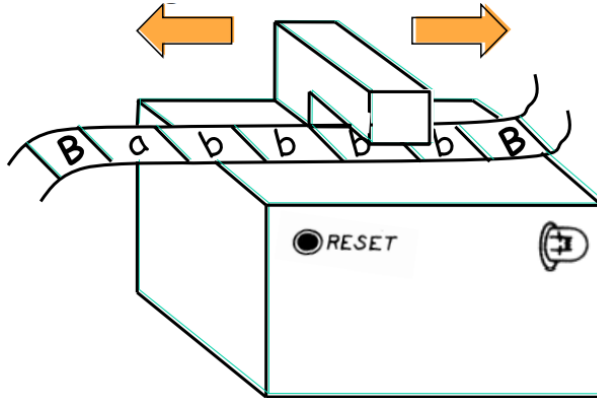






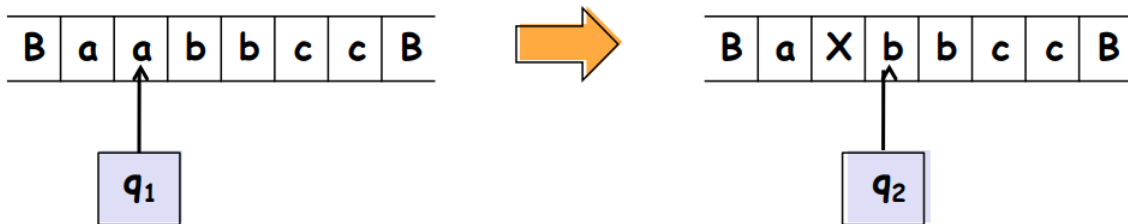


The tape contains a blank symbol, represented by B , which initially fills the entire tape

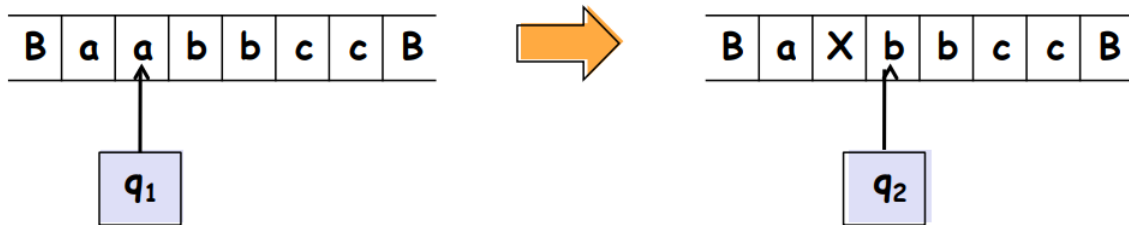


*The input string is placed on the tape;
the remaining symbols are blank B*

Transitions

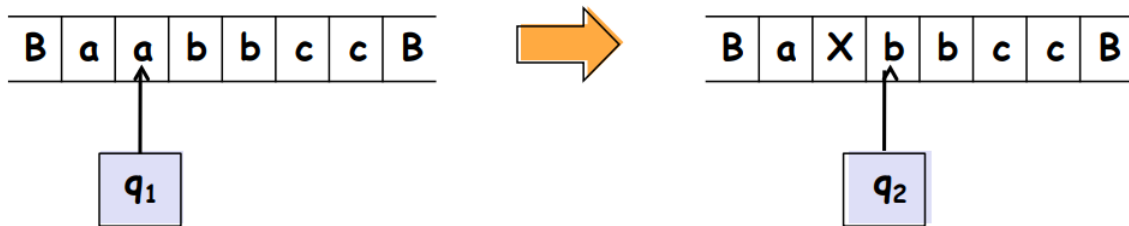


Transitions



*The symbol a is replaced with X
and the head moves to the right*

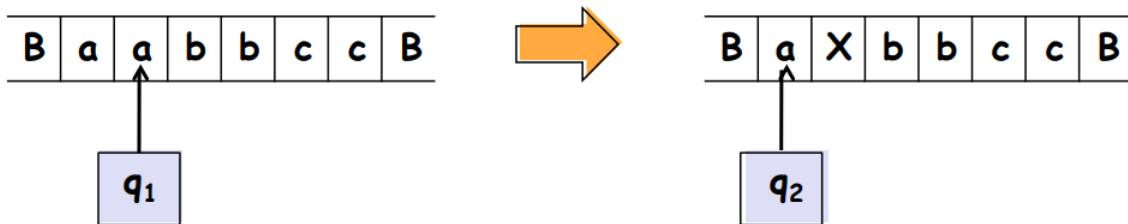
Transitions



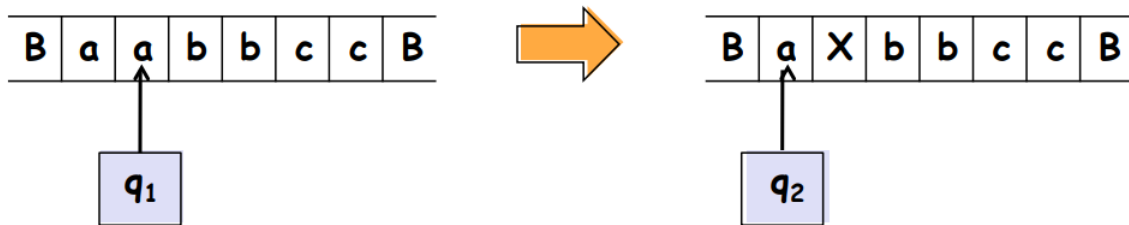
$$\delta(q_1, a) = (q_2, X, R)$$

*In state q_1 , the machine reads a from the tape,
writes X in its place, and moves to the right*

Transitions

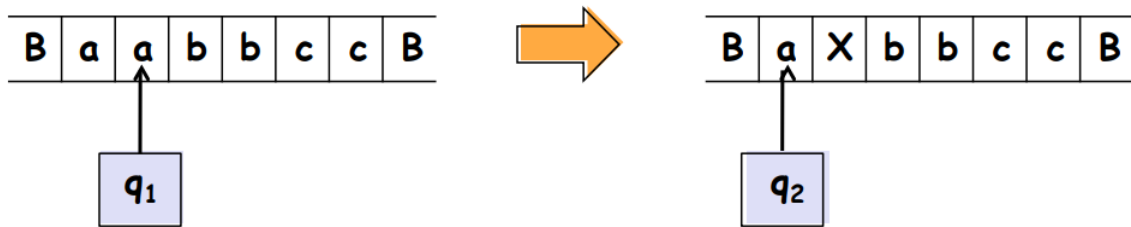


Transitions



*The symbol a is replaced with X
and the head moves to the left*

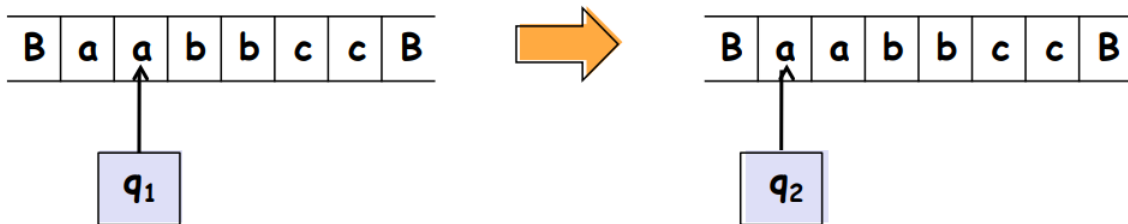
Transitions



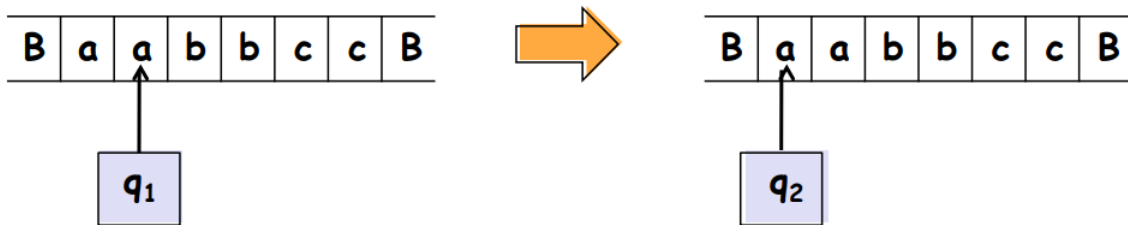
$$\delta(q_1, a) = (q_2, X, L)$$

*In state q_1 , the machine reads a from the tape,
writes X , and moves to the left*

Transitions

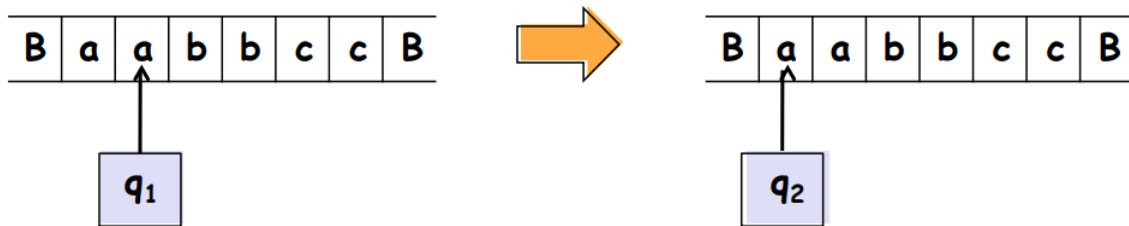


Transitions



*The symbol **a** remains unchanged
and the head moves to the left*

Transitions



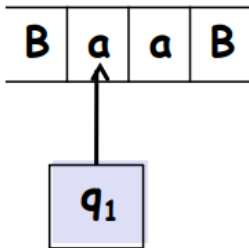
$$\delta(q_1, a) = (q_2, a, L)$$

*In state q_1 , the machine reads a from the tape,
writes a again, and moves to the left*

- Determine the final state of the tape considering the following transitions:

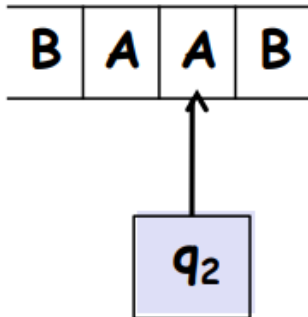
$$\delta(q_1, a) = (q_1, A, R)$$

$$\delta(q_1, B) = (q_2, B, L)$$



$$\delta(q_1, a) = (q_1, A, R)$$

$$\delta(q_1, B) = (q_2, B, L)$$

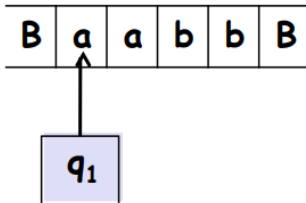


- Determine the final state of the tape considering the following transitions:

$$\delta(q_1, a) = (q_1, d, R)$$

$$\delta(q_1, b) = (q_1, d, R)$$

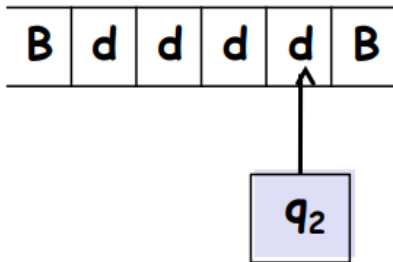
$$\delta(q_1, B) = (q_2, B, L)$$



$$\delta(q_1, a) = (q_1, d, R)$$

$$\delta(q_1, b) = (q_1, d, R)$$

$$\delta(q_1, B) = (q_2, B, L)$$



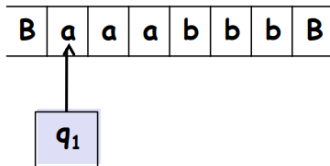
- Determine the final state of the tape considering the following transitions:

$$\delta(q_1, a) = (q_1, a, R)$$

$$\delta(q_1, b) = (q_1, a, R)$$

$$\delta(q_1, B) = (q_2, B, L)$$

$$\delta(q_2, a) = (q_2, d, L)$$

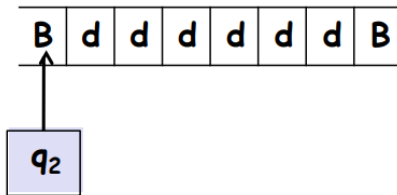


$$\delta(q_1, a) = (q_1, a, R)$$

$$\delta(q_1, b) = (q_1, a, R)$$

$$\delta(q_1, B) = (q_2, B, L)$$

$$\delta(q_2, a) = (q_2, d, L)$$



- Determine the final state of the tape considering the following transitions:

$$\delta(q_1, a) = (q_2, c, R)$$

$$\delta(q_2, a) = (q_2, a, R)$$

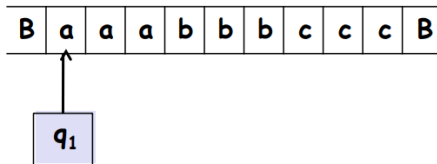
$$\delta(q_2, b) = (q_3, d, L)$$

$$\delta(q_2, d) = (q_2, d, R)$$

$$\delta(q_3, d) = (q_3, d, L)$$

$$\delta(q_3, a) = (q_3, a, L)$$

$$\delta(q_3, c) = (q_1, c, R)$$



$$\delta(q_1, a) = (q_2, c, R)$$

$$\delta(q_2, a) = (q_2, a, R)$$

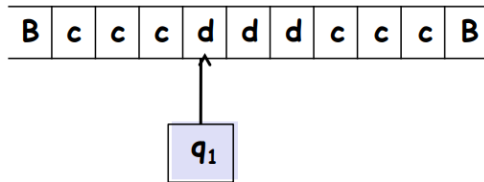
$$\delta(q_2, b) = (q_3, d, L)$$

$$\delta(q_2, d) = (q_2, d, R)$$

$$\delta(q_3, d) = (q_3, d, L)$$

$$\delta(q_3, a) = (q_3, a, L)$$

$$\delta(q_3, c) = (q_1, c, R)$$



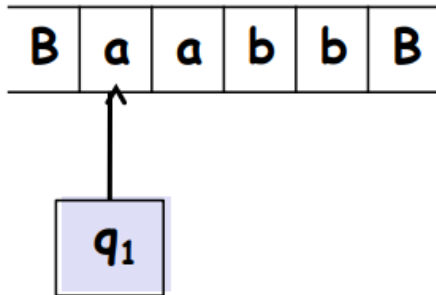
- Determine the final state of the tape considering the following transitions:

$$\delta(q_1, a) = (q_2, a, R)$$

$$\delta(q_1, b) = (q_1, b, R)$$

$$\delta(q_2, a) = (q_1, a, L)$$

$$\delta(q_2, B) = (q_3, B, L)$$

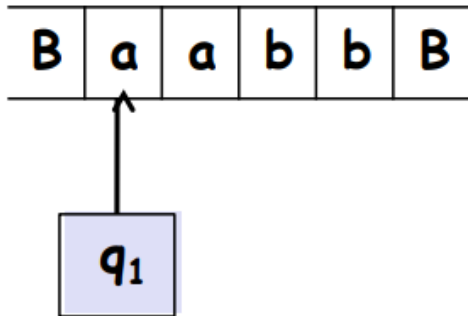


$$\delta(q_1, a) = (q_2, a, R)$$

$$\delta(q_1, b) = (q_1, b, R)$$

$$\delta(q_2, a) = (q_1, a, L)$$

$$\delta(q_2, B) = (q_3, B, L)$$



*The TM does not halt.
It is said to remain in an infinite loop*

Turing Machine

A Turing machine is a collection of 7 elements:

- Q : set of states
- Σ : input alphabet
- Γ : tape alphabet
- $q \in Q$: initial state
- $B \in \Gamma$: blank symbol initially written across the entire tape
- $F \subseteq Q$: set of final states
- $\delta : Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$: transition relation

String acceptance

A string w is accepted by a Turing machine M if the computation that starts in (q_1, w) ends in an acceptance state

String acceptance

A string w is accepted by a Turing machine M if the computation that starts in (q_1, w) ends in an acceptance state

- No condition is imposed on the final configuration of the tape
- It is not required that the entire string be consumed

Transition Diagram

It allows the formal definition of a Turing machine to be represented graphically

- $Q = \{q_1, q_2, q_3\}$: set of states
- $\Sigma = \{a, b\}$
- $\Gamma = \{B, a, b, d\}$
- q_1 is the initial state
- B is the blank symbol
- $F = \{q_3\}$
- $\delta : Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$ is the transition relation

$$\delta(q_1, a) = (q_1, a, R)$$

$$\delta(q_1, b) = (q_1, a, R)$$

$$\delta(q_1, B) = (q_2, B, L)$$

$$\delta(q_2, a) = (q_2, d, L)$$

$$\delta(q_2, d) = (q_3, d, L)$$

$$\delta(q_2, B) = (q_2, B, R)$$

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- q_1 is the initial state
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- $\delta : Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$



q₁



q₂



q₃

$$\delta(q_1, a) = (q_1, a, R)$$

$$\delta(q_1, b) = (q_1, a, R)$$

$$\delta(q_1, B) = (q_2, B, L)$$

$$\delta(q_2, a) = (q_2, d, L)$$

$$\delta(q_2, d) = (q_3, d, L)$$

$$\delta(q_2, B) = (q_2, B, R)$$

$$\delta(q_1, a) = (q_1, a, R)$$

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$$\delta(q_2, a) = (q_2, d, L)$$

$$\delta(q_2, d) = (q_3, d, L)$$

$$\delta(q_2, B) = (q_2, B, R)$$



q₁



q₂



q₃

$$\delta(q_1, a) = (q_1, a, R)$$

$$\delta(q_1, b) = (q_1, a, R)$$

$$\delta(q_1, B) = (q_2, B, L)$$

$$\delta(q_2, a) = (q_2, d, L)$$

$$\delta(q_2, d) = (q_3, d, L)$$

$$\delta(q_2, B) = (q_2, B, R)$$



$$\delta(q_1, a) = (q_1, a, R)$$

$$\delta(q_1, b) = (q_1, a, R)$$

$$\delta(q_1, B) = (q_2, B, L)$$

$$\delta(q_2, a) = (q_2, d, L)$$

$$\delta(q_2, d) = (q_3, d, L)$$

$$\delta(q_2, B) = (q_2, B, R)$$

a/a,R



q₁



q₂



q₃

$$\delta(q_1, a) = (q_1, a, R)$$

$$\delta(q_1, b) = (q_1, a, R)$$

$$\delta(q_1, B) = (q_2, B, L)$$

$$\delta(q_2, a) = (q_2, d, L)$$

$$\delta(q_2, d) = (q_3, d, L)$$

$$\delta(q_2, B) = (q_2, B, R)$$

a/a,R

b/a,R



q₁



q₂



q₃

$$\delta(q_1, a) = (q_1, a, R)$$

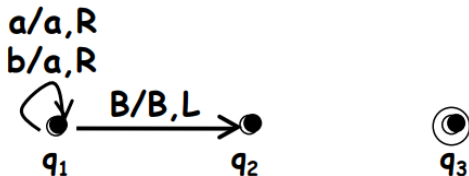
$$\delta(q_1, b) = (q_1, a, R)$$

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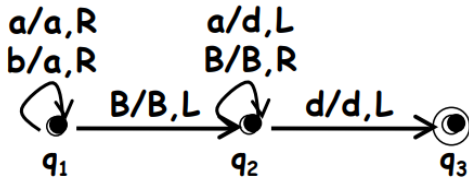
$$\delta(q_2, a) = (q_2, d, L)$$

$$\delta(q_2, d) = (q_3, d, L)$$

$$\delta(q_2, B) = (q_2, B, R)$$



$$\begin{aligned}\delta(q_1, a) &= (q_1, a, R) \\ \delta(q_1, b) &= (q_1, a, R) \\ \delta(q_1, B) &= (q_2, B, L) \\ \delta(q_2, a) &= (q_2, d, L) \\ \delta(q_2, d) &= (q_3, d, L) \\ \delta(q_2, B) &= (q_2, B, R)\end{aligned}$$



1. Turing Machines

2. Designing Turing Machines

Construct a TM for the following languages:

- Regular Languages (RL)

$$a^*, a^+b^+, (a \cup b)^*aa(a \cup b)^*$$

- Context-Free Languages (CFL)

$$a^n b^n, a^n b^{2n}, a^m b^n c^{m+n}, a^m b^n c^{m-n}$$

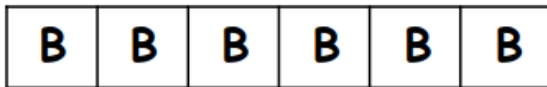
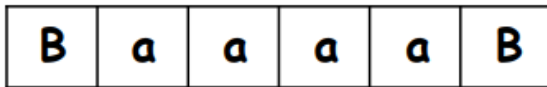
$$a^{3n} b^n, a^n b^m c^n, a^{2n} b^m c^{2m} d^n, a^n b^{n+m} c^m$$

- Recursively Enumerable Languages (RE)

$$a^n b^n c^n, wcw, a^n b^m c^n d^m, a^n b^n c^m \ (m > n)$$

Construct a TM that accepts a^*

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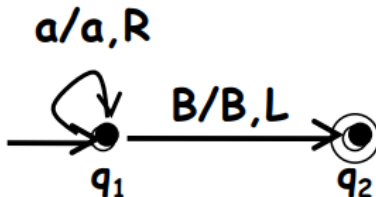


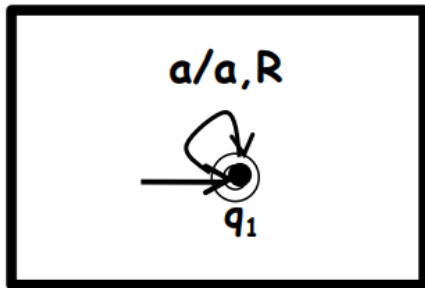
Construct a TM that accepts a^*

Idea: move along the tape for each a until reaching B .
Then transition to an acceptance state

Construct a TM that accepts a^*

Idea: move along the tape for each a until reaching B .
Then transition to an acceptance state

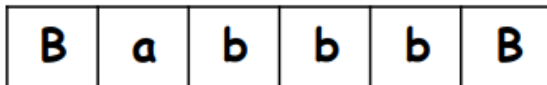
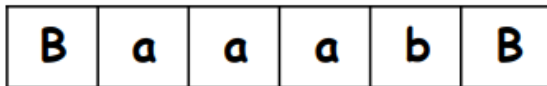




Why is this TM not valid for accepting a^* ?

Construct a TM that accepts a^+b^+

Construct a TM that accepts a^+b^+

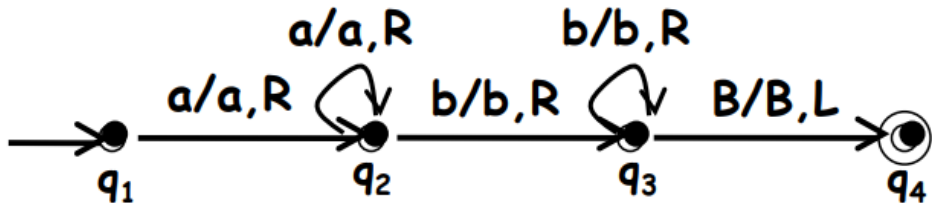


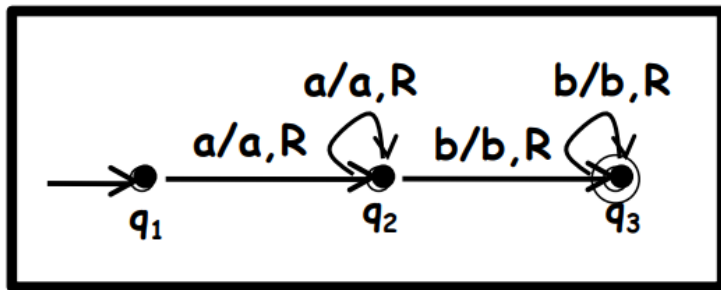
Construct a TM that accepts a^+b^+

Idea: move along the tape for each a until reaching a b.
Then continue moving through each b until finding a B

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Why is this TM not valid for accepting a^+b^+ ?

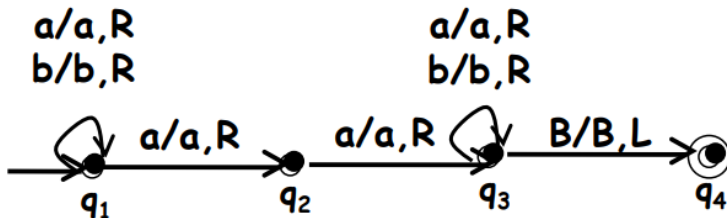
Construct a TM that accepts $(a \cup b)^*aa(a \cup b)^*$

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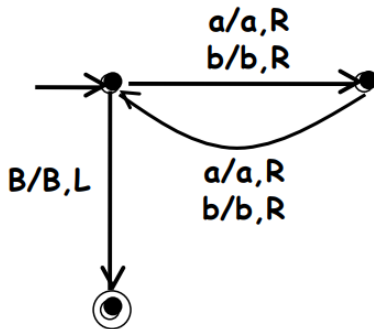
Construct a TM that accepts

$$L = \{w \in \{a, b\}^* \mid w \text{ has even length}\}$$

Idea: for each a or b, the symbol remains unchanged.
The machine reaches an acceptance state only if the symbols are processed in pairs

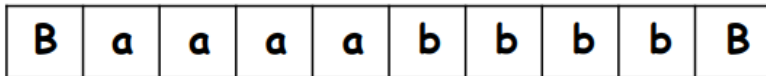
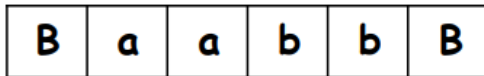
Idea: for each a or b, the symbol remains unchanged.

The machine reaches an acceptance state only if the symbols are processed in pairs



Construct a TM that accepts $L = \{a^n b^n, n \geq 1\}$

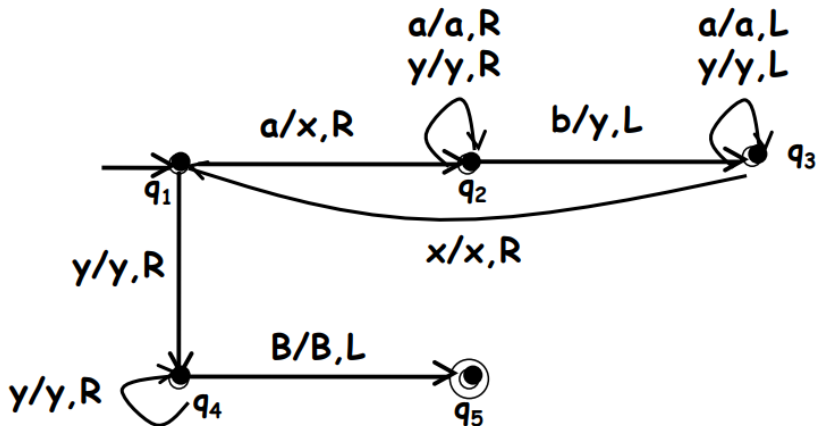
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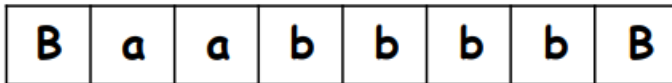
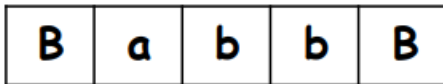
Idea: start with the leftmost a and convert it into c, then move to the right until finding the first b and replace it with d. Then move to the left until finding a c and repeat the process. The machine transitions to an acceptance state only if no a's or b's remain

TM that accepts $L = \{a^n b^n, n \geq 1\}$



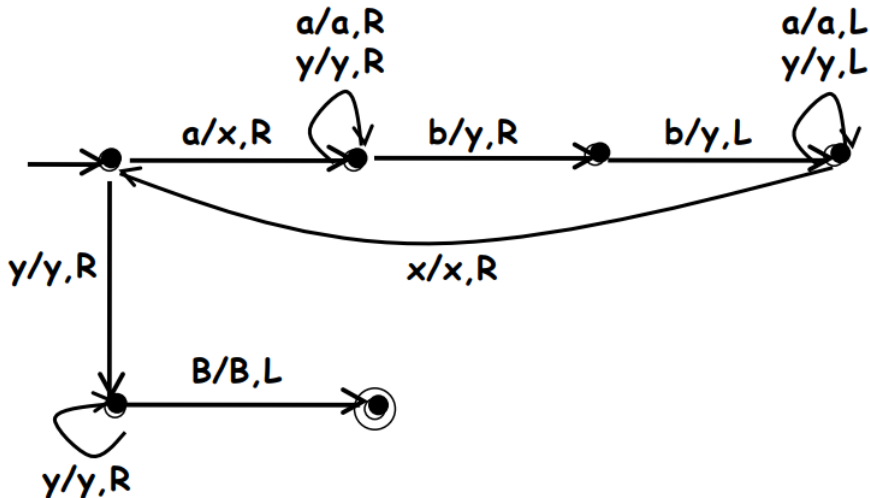
Construct a TM that accepts $L = \{a^n b^{2^n}, n \geq 1\}$

Construct a TM that accepts $L = \{a^n b^{2n}, n \geq 1\}$



Construct a TM that accepts $L = \{a^n b^{2n}, n \geq 1\}$

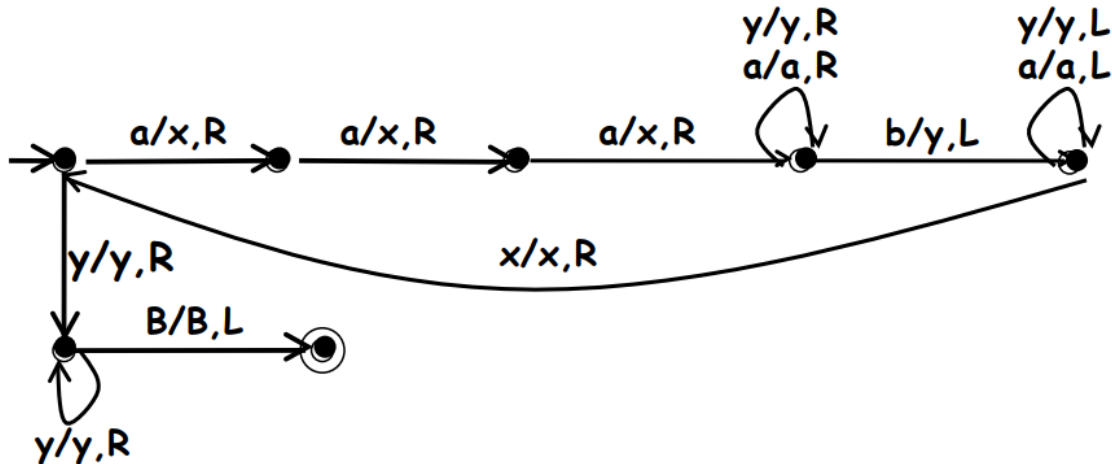
Idea: for each a, an x is written. Then two consecutive b's are searched and replaced with y. The machine moves to the left searching for the x and repeats the process for every a



Construct a TM that accepts $L = \{a^{3n}b^n, n \geq 1\}$

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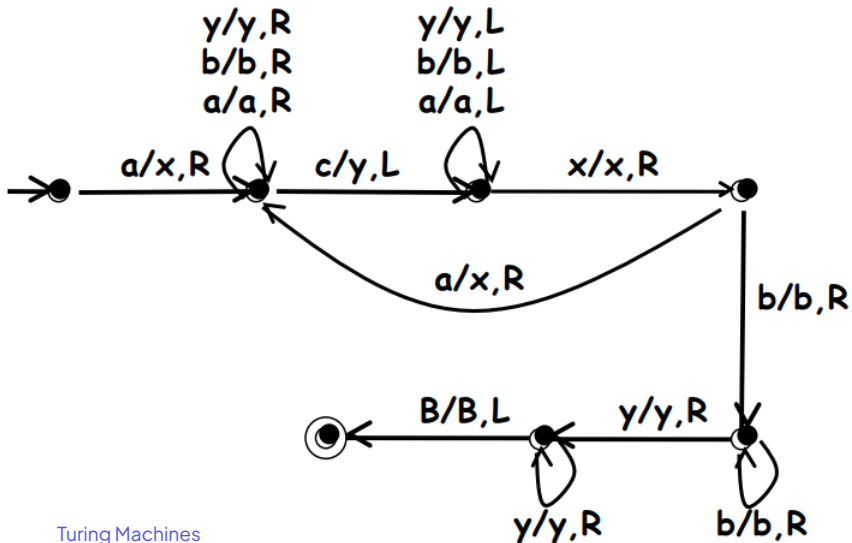
Idea: three a's are replaced with x, and then the machine searches for a b, which is converted into y. The machine moves to the left searching for the x and repeats the process



Construct a TM that accepts $L = \{a^n b^m c^n, m, n \geq 1\}$

Construct a TM that accepts $L = \{a^n b^m c^n, m, n \geq 1\}$

Idea: each a is replaced with x , and the machine moves through any number of b 's searching for a c , which is replaced with y . The machine then moves to the left searching for the x and repeats the process



Construct a TM that accepts $L = \{a^m b^n c^{m+n}, m, n \geq 1\}$

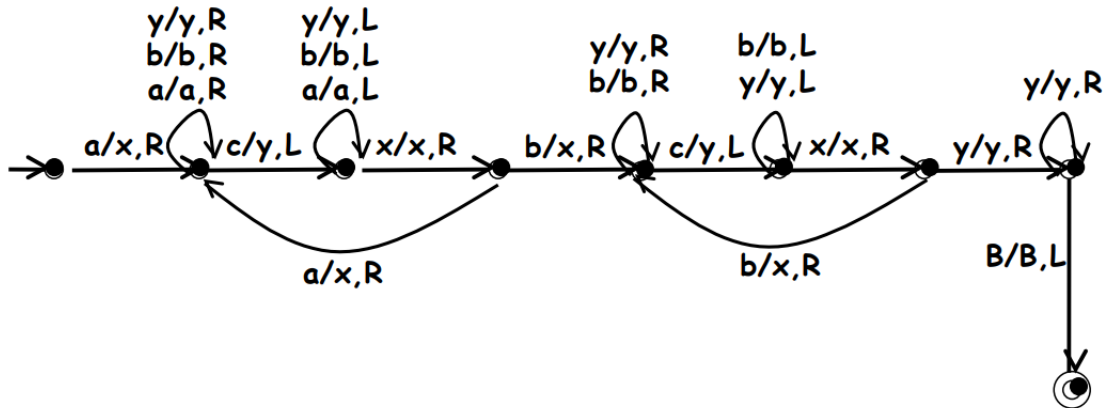
Construct a TM that accepts $L = \{a^m b^n c^{m+n}, m, n \geq 1\}$

B	a	b	b	c	c	c	B
---	---	---	---	---	---	---	---

B	a	a	b	b	b	c	c	c	c	c	B
---	---	---	---	---	---	---	---	---	---	---	---

Construct a TM that accepts $L = \{a^m b^n c^{m+n}, m, n \geq 1\}$

Idea: for each a, an x is written and a c is searched, which is replaced with y. The machine moves to the left searching for the x and repeats the process for every a. When a b is read, it is replaced with x and a c is searched, which is replaced with y. Then the machine moves to the left searching for an x and repeats the process



Construct a TM that accepts $L = \{a^{2n}b^m c^{2m}d^n, m, n \geq 1\}$

Construct a TM that accepts $L = \{a^{2n}b^m c^{2m}d^n, m, n \geq 1\}$

Idea: two a's are replaced with x and a d is searched, which is replaced with y. The machine moves to the left until finding the x and repeats the procedure. If after the x there is a b, it is replaced with x and two c's are searched, which are replaced with y's. The machine moves to the left searching for an x and repeats the procedure

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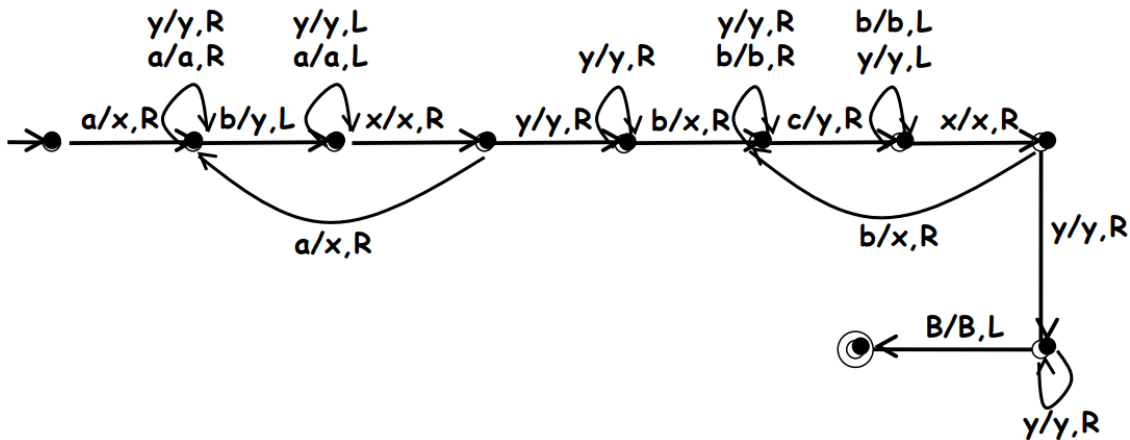


Construct a TM that accepts $L = \{a^n b^{n+m} c^m, m, n \geq 1\}$

Construct a TM that accepts $L = \{a^n b^{n+m} c^m, m, n \geq 1\}$

Idea: for each a, it is replaced with x and a b is searched, which is replaced with y. The machine moves to the left until finding an x and repeats the procedure. When all the a's are finished, each x is replaced with y and a c is searched, which is replaced with y. The machine moves to the left and repeats the procedure

Turing Machines



Analyze the possibility of designing a TM that accepts each of the following languages:

- $\{a^n b^n c^n \mid n \geq 1\}$
- $\{wcw \mid w \in \{a, b\}^*\}$
- $\{a^n b^m c^n d^m \mid n, m \geq 1\}$
- $\{a^n b^n c^m \mid m \geq n\}$

TM that accepts $L = \{a^n b^n c^n, n \geq 1\}$

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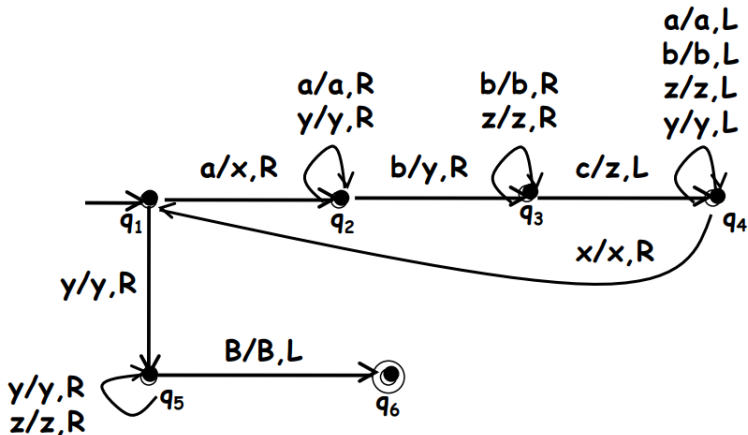
B	a	a	b	b	c	c	B
---	---	---	---	---	---	---	---

B	a	a	a	b	b	b	c	c	c	B
---	---	---	---	---	---	---	---	---	---	---

TM that accepts $L = \{a^n b^n c^n, n \geq 1\}$

Idea: start with the leftmost a and convert it into x, move to the right until finding the first b and replace it with y, then move to the right until finding a c, which is replaced with z. Move to the left until finding an x, then move to the right and repeat the process

TM that accepts $L = \{a^n b^n c^n, n \geq 1\}$



TM that accepts $L = \{wcw \mid w \in \{a, b\}^*\}$

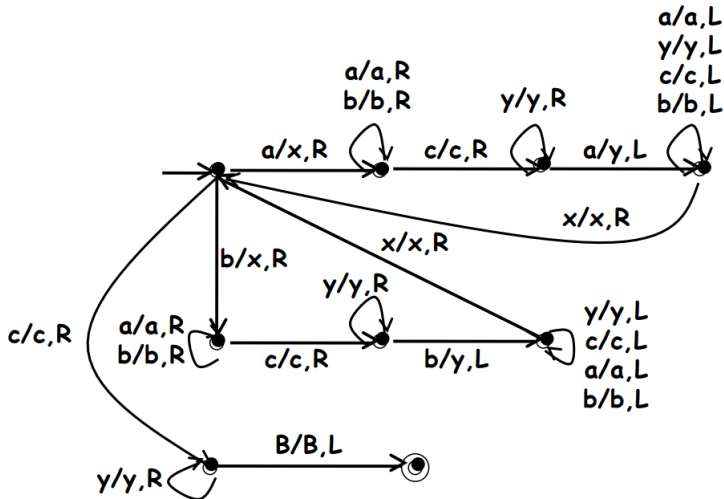
TM that accepts $L = \{wcw \mid w \in \{a, b\}^*\}$

B	a	b	b	c	a	b	b	B
---	---	---	---	---	---	---	---	---

B	b	a	a	b	c	b	a	a	b	B
---	---	---	---	---	---	---	---	---	---	---

TM that accepts $L = \{wcw \mid w \in \{a, b\}^*\}$

Idea: take the initial symbol, which can be a or b, replace it with x, and move to the right until finding the c. Then search for the same symbol that was replaced with x. Move to the left until finding the x and repeat the process



TM that accepts $L = \{a^n b^m c^n d^m, m, n \geq 1\}$

TM that accepts $L = \{a^n b^m c^n d^m, m, n \geq 1\}$

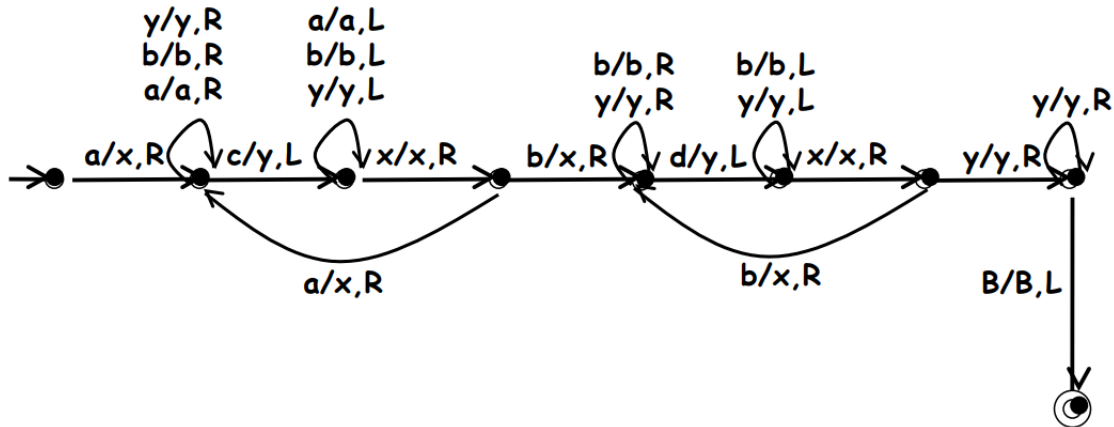
B	a	b	b	c	d	d	B
---	---	---	---	---	---	---	---

B	a	a	b	b	b	c	c	d	d	d	B
---	---	---	---	---	---	---	---	---	---	---	---

TM that accepts $L = \{a^n b^m c^n d^m, m, n \geq 1\}$

Idea: each a is replaced with x and a c is searched, which is changed to y . The machine moves to the left until finding an x and repeats the procedure. When all the a 's are finished, a b is replaced with x and a d is searched, which is changed to y . The machine moves to the left until finding an x and repeats the procedure

Turing Machines





Kozen, D. C. (2007)

Automata and computability

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